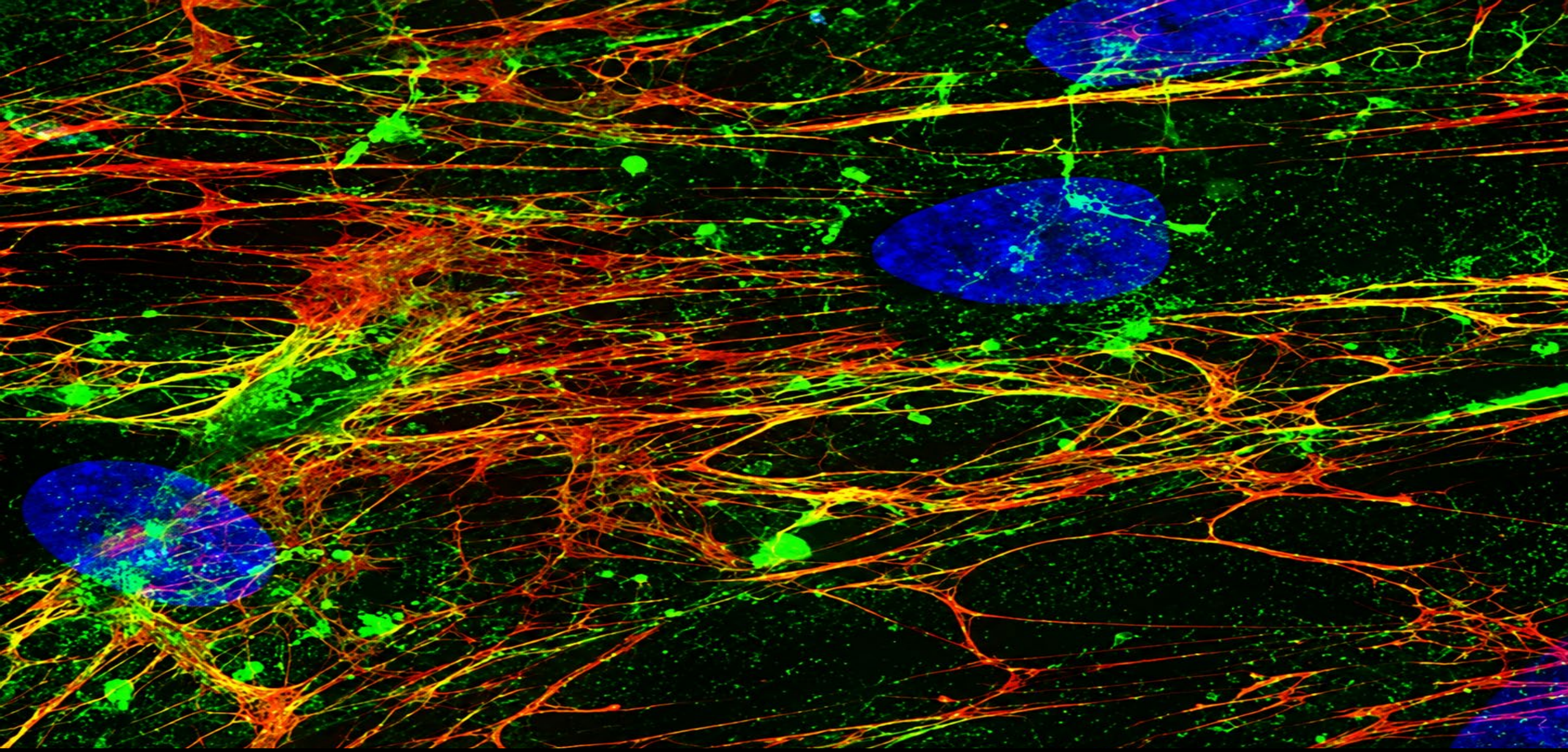
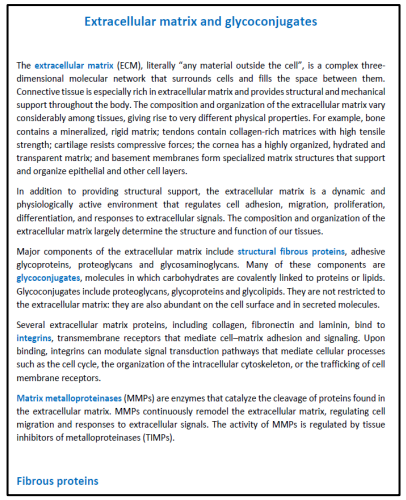


Extracellular matrix and glycoconjugates

Molecular components and functions

Supportive material



Extracellular matrix and glycoconjugates

The extracellular matrix (ECM), literally "any material outside the cell", is a complex three-dimensional molecular network that surrounds cells and fills the space between them. Connective tissue is especially rich in extracellular matrix and provides structural and mechanical support throughout the body. The composition and organization of the extracellular matrix vary considerably among tissues, giving rise to very different physical properties. For example, bone contains a mineralized, rigid matrix; tendons contain collagen-rich matrices with high tensile strength; cartilage resists compressive forces; the cornea has a highly organized, hydrated and transparent matrix; and basement membranes form specialized matrix structures that support and organize epithelial and other cell layers.

In addition to providing structural support, the extracellular matrix is a dynamic and physiologically active environment that regulates cell adhesion, migration, proliferation, differentiation, and responses to extracellular signals. The composition and organization of the extracellular matrix largely determine the structure and function of our tissues.

Major components of the extracellular matrix include structural fibrous proteins, adhesive glycoproteins, proteoglycans and glycosaminoglycans. Many of these components are glycoconjugates, molecules in which carbohydrates are covalently linked to proteins or lipids. Glycoconjugates include proteoglycans, glycoproteins and glycolipids. They are not restricted to the extracellular matrix: they are also abundant on the cell surface and in secreted molecules.

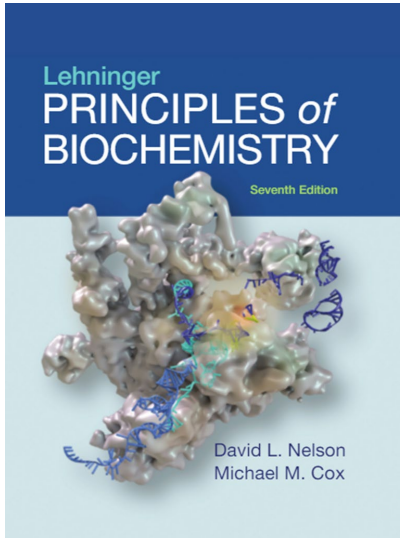
Several extracellular matrix proteins, including collagen, fibronectin and laminin, bind to integrins, transmembrane receptors that mediate cell–matrix adhesion and signaling. Upon binding, integrins can modulate signal transduction pathways that mediate cellular processes such as the cell cycle, the organization of the intracellular cytoskeleton, or the trafficking of cell membrane receptors.

Matrix metalloproteinases (MMPs) are enzymes that catalyze the cleavage of proteins found in the extracellular matrix. MMPs continuously remodel the extracellular matrix, regulating cell migration and responses to extracellular signals. The activity of MMPs is regulated by tissue inhibitors of metalloproteinases (TIMPs).

Fibrous proteins

Additional study material

Aula Virtual document: ECM and glycoconjugates



Reading

Lehninger's PoB Chapter 4: 4.3. Fibrous proteins

Lehninger's PoB Chapter 7: 7.3: Glycoconjugates

Learning goals

- Definition and function of the extracellular matrix
- Fibrous proteins: concept and examples
- Basic structure and biological function of selected fibrous proteins: α-keratin and collagen
- Types of glycoconjugates
- Biological importance of the glycosylation of proteins
- O- and N-glycosylation of proteins

Extracellular matrix

- Extracellular material organized as a three-dimensional network surrounding cells
- Provides structural and biochemical support to surrounding cells

  strength, stiffness, elasticity, and physical barrier in interstitial matrices and basement membranes
- Main components:
  - Structural fibrous proteins

    collagen, elastin
  - Adhesive glycoproteins

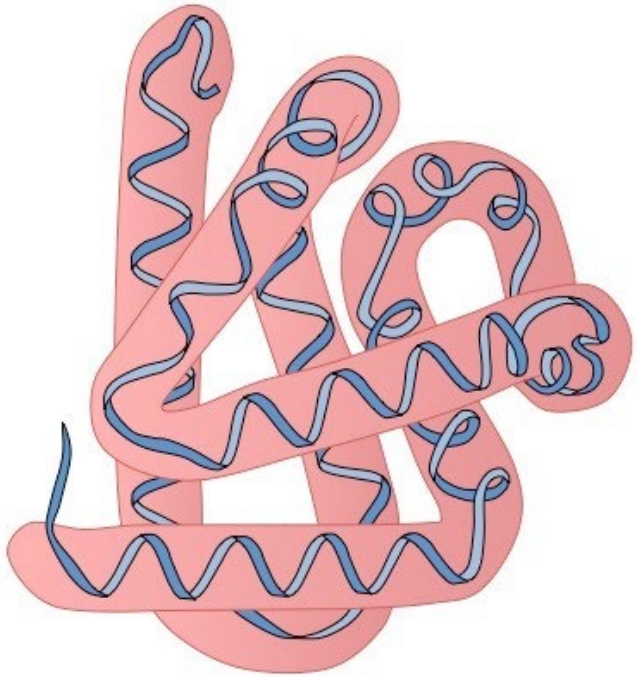
    fibronectin, laminin
  - Proteoglycans and glycosaminoglycans

Fibrous proteins

- Polypeptide chains arranged in long strands or sheets
- Dominated by repetitive secondary structures
- Provide support, shape and external protection



Fibrous Protein



Globular Protein

α-keratin

- Two α-helices wrapped about each other in a left-handed sense and cross-linked by disulfide bonds
- Rich in hydrophobic amino acids

  Ala, Val, Leu, Ile, Met, Phe

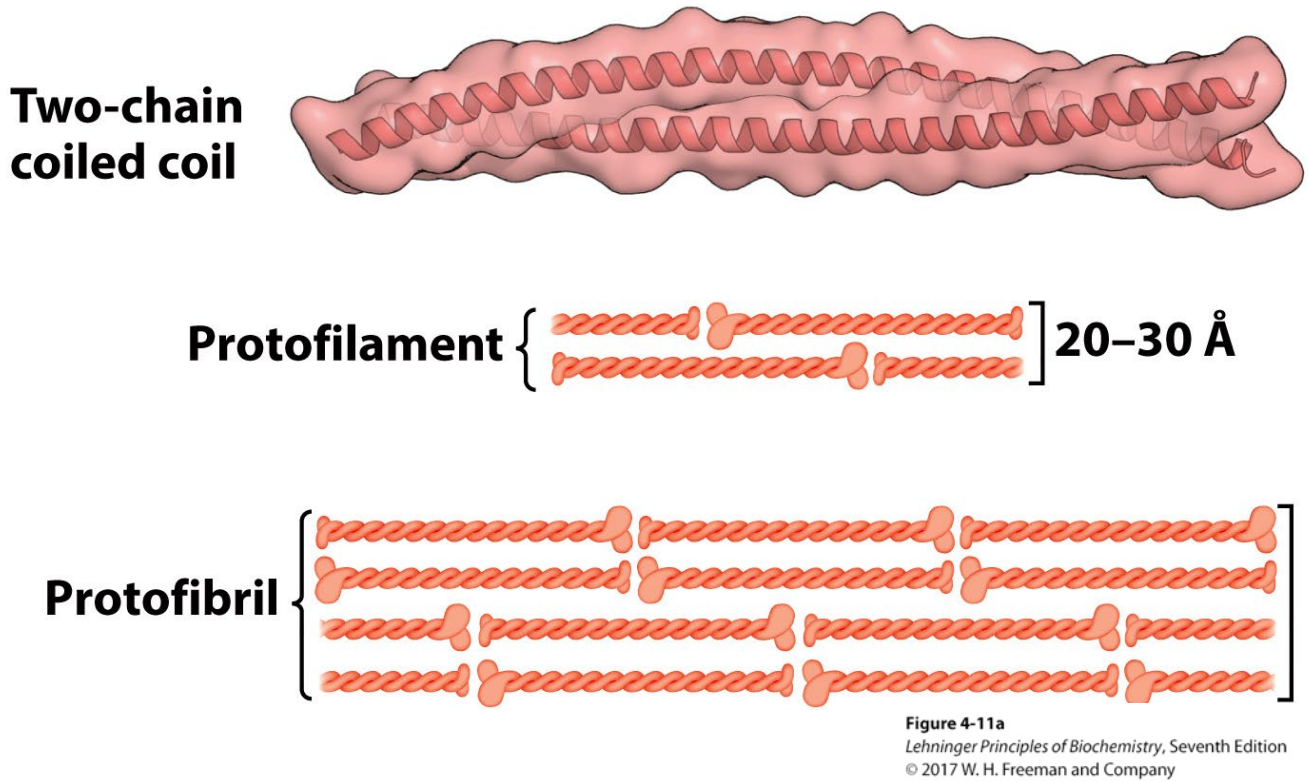


Figure 4-11a
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- The disulfide bridges resist the forces that try to deform the keratin fibrils
- Soft keratin (in the epidermis) has low sulfur content (fewer disulfide bridges). Hard keratin (in hair and nails) has high sulfur content

Collagen

- Most abundant protein in human body
- Major component of connective tissues

  tendons, cartilage, bones, cornea

- Each collagen chain is a long left-handed helix of a repeating tripeptide unit Gly-X-Y, where X is often Pro and Y is often 4-hydroxyproline
- Three collagen chains intertwine into a right-handed triple helix
- Type I collagen has a tensile strength comparable to or greater than a steel wire of equal cross section

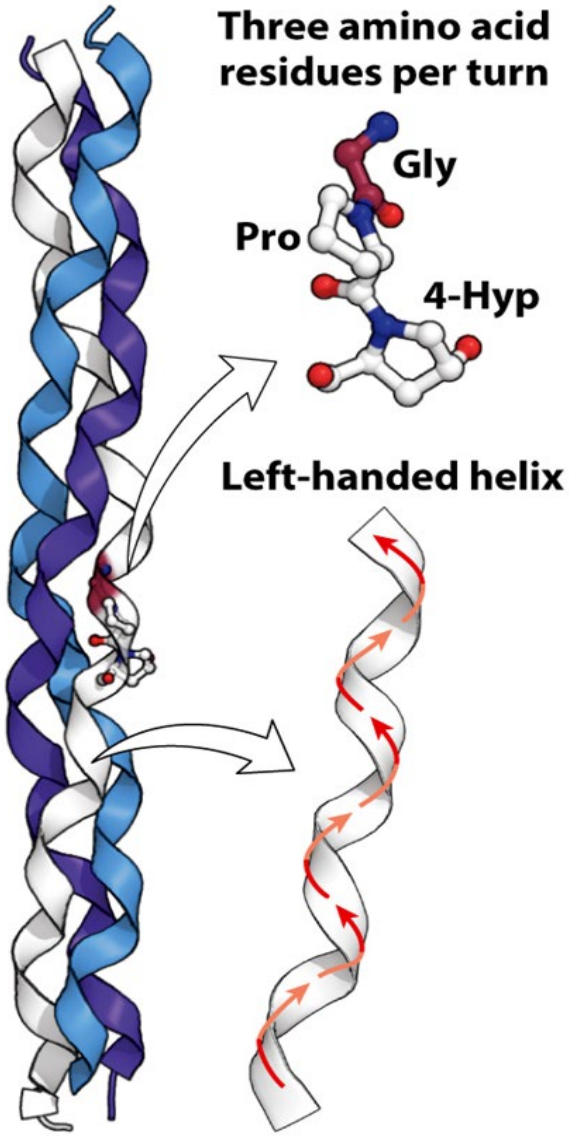
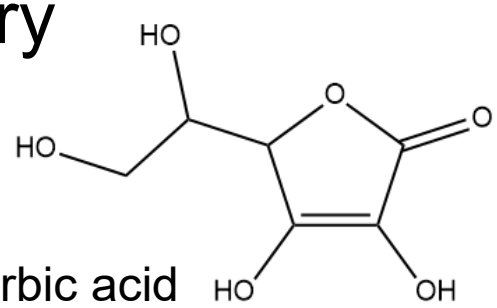


Figure 4-12
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4-Hydroxyproline

- Helps stabilize the collagen triple helix
- Increases the stability of collagen
- Hydroxylation is catalyzed by prolyl hydroxylase
- Prolyl hydroxylase requires ascorbate (vitamin C)
- Citrus fruits have high levels of vitamin C
- Vitamin C was identified in the 20th century



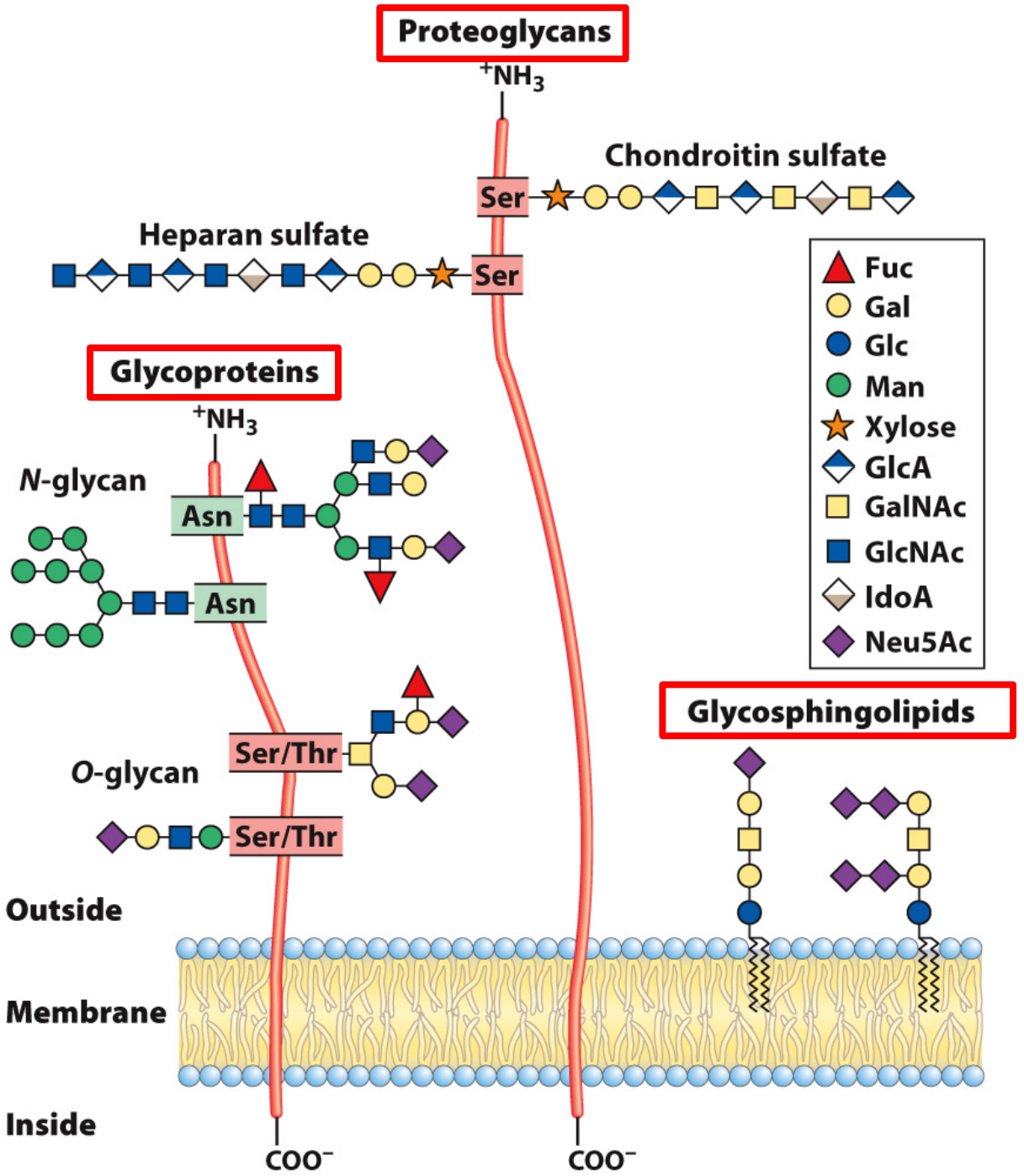
Ascorbic acid

Collagen-associated diseases

- Different collagens are expressed in different tissues, so mutations produce very different diseases
- Some of the most clinically relevant:
    - Type I Osteogenesis imperfecta
    - Type III Vascular Ehlers-Danlos syndrome
    - Type IV Alport's syndrome
    - Type VII Dystrophic epidermolysis bullosa

Glycoconjugates

- Proteoglycans
- Glycoproteins
- Glycolipids



Proteoglycans

- Major components of the ECM
- A core protein with one or more covalently attached glycosaminoglycan chains, usually through Ser residues
- Regulate adhesion, movement of cells, activity and stability of proteins, signaling, etc.
- Some are attached to cell membranes and others are secreted into the ECM

Membrane proteoglycans

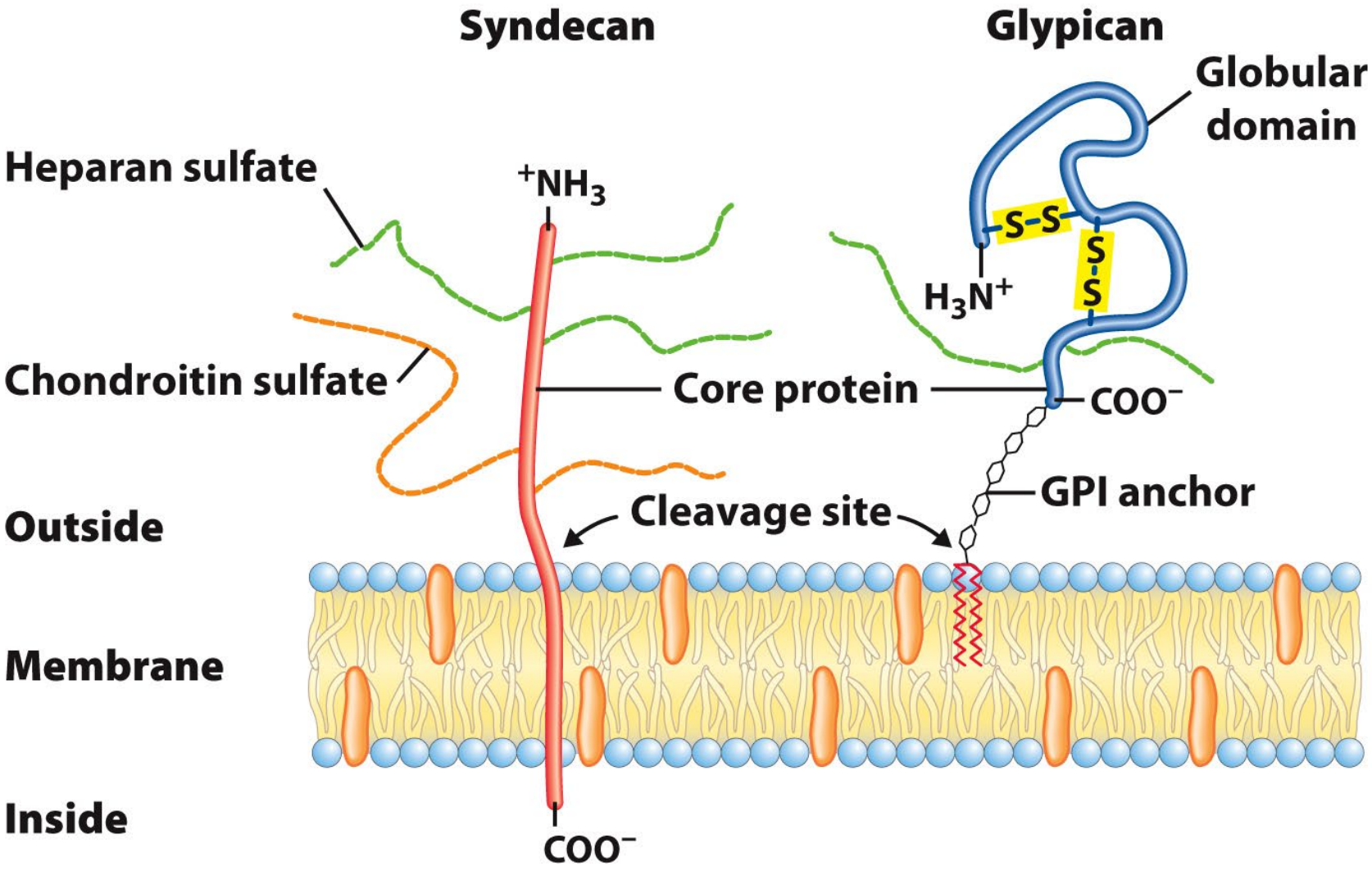


Figure 7-25a
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Proteoglycan aggregates

- Supramolecular assemblies of many proteoglycans around a hyaluronan
- Proteoglycan aggregates of aggrecan provide hydration, lubrication, resistance to compression, and resilience of cartilage

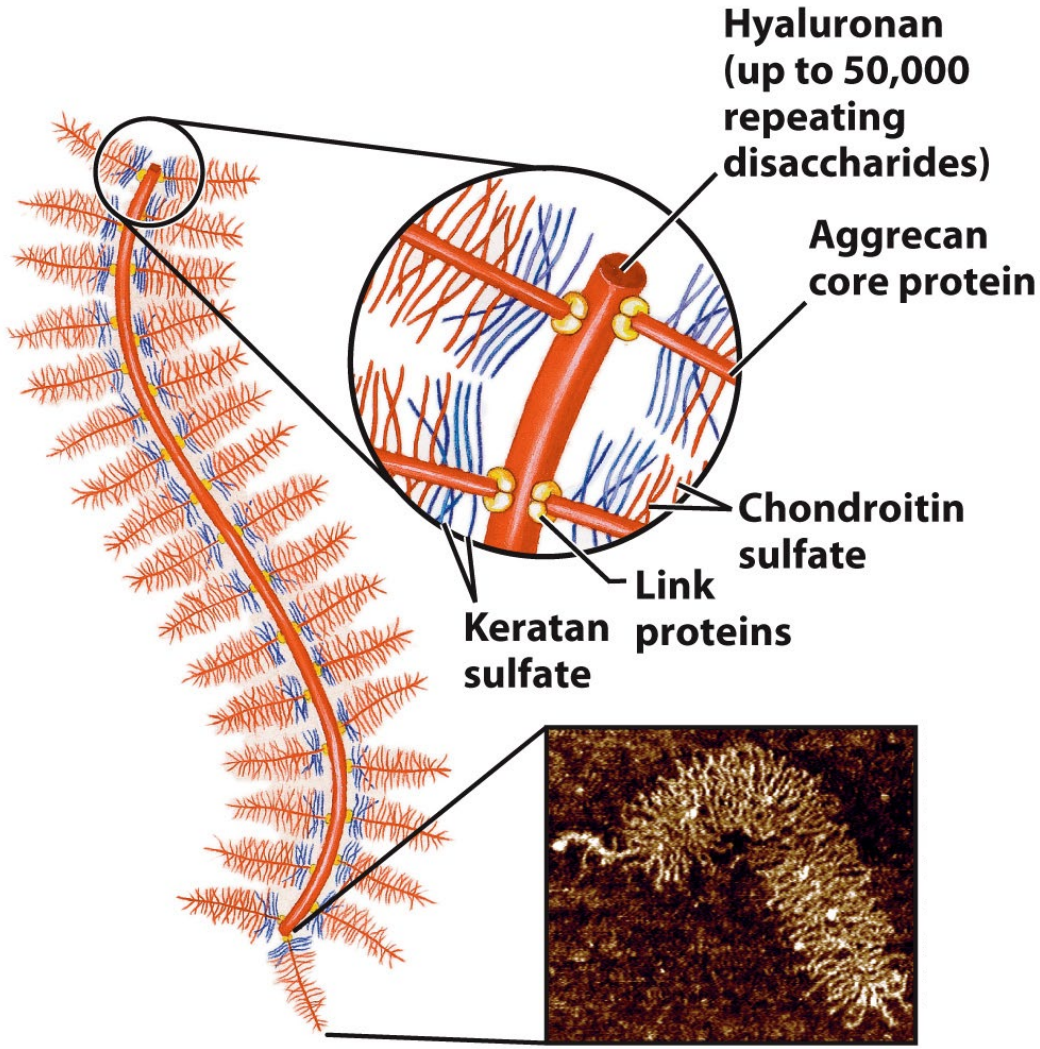


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Cell-matrix interactions

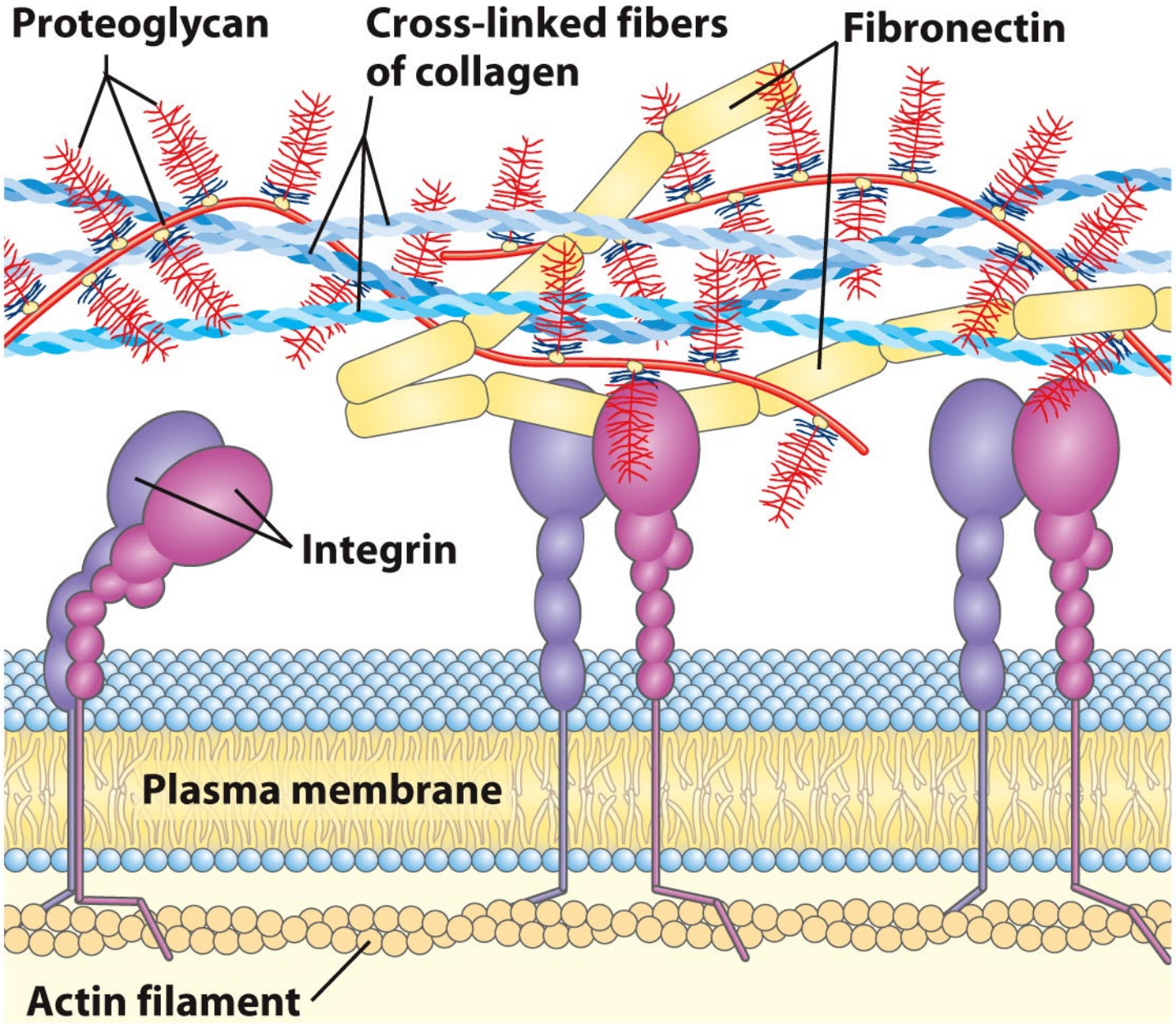


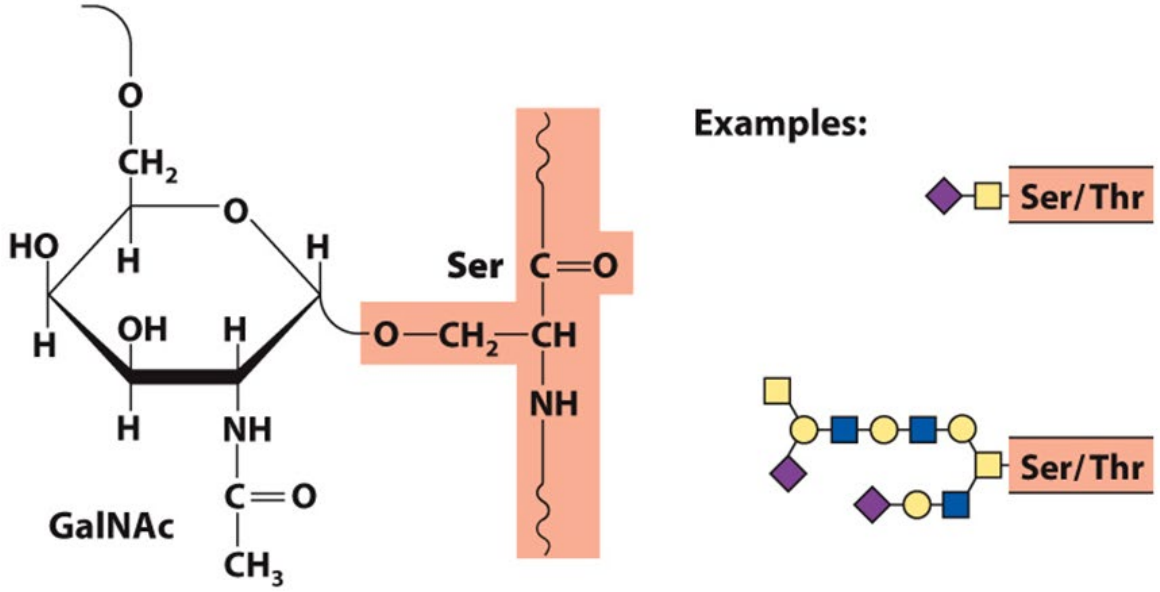
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Glycoproteins

- A protein with small oligosaccharides attached
  - The carbohydrate is attached via its anomeric carbon to amino acids on the protein
  - More than half of human proteins are glycoproteins
- Carbohydrates play important roles in molecular recognition and protein function
  - Recognition by enzymes
  - Receptor binding
  - Immunological recognition
  - Cell-cell adhesion
  - Protein trafficking

Glycosylation of proteins

O-linked



N-linked

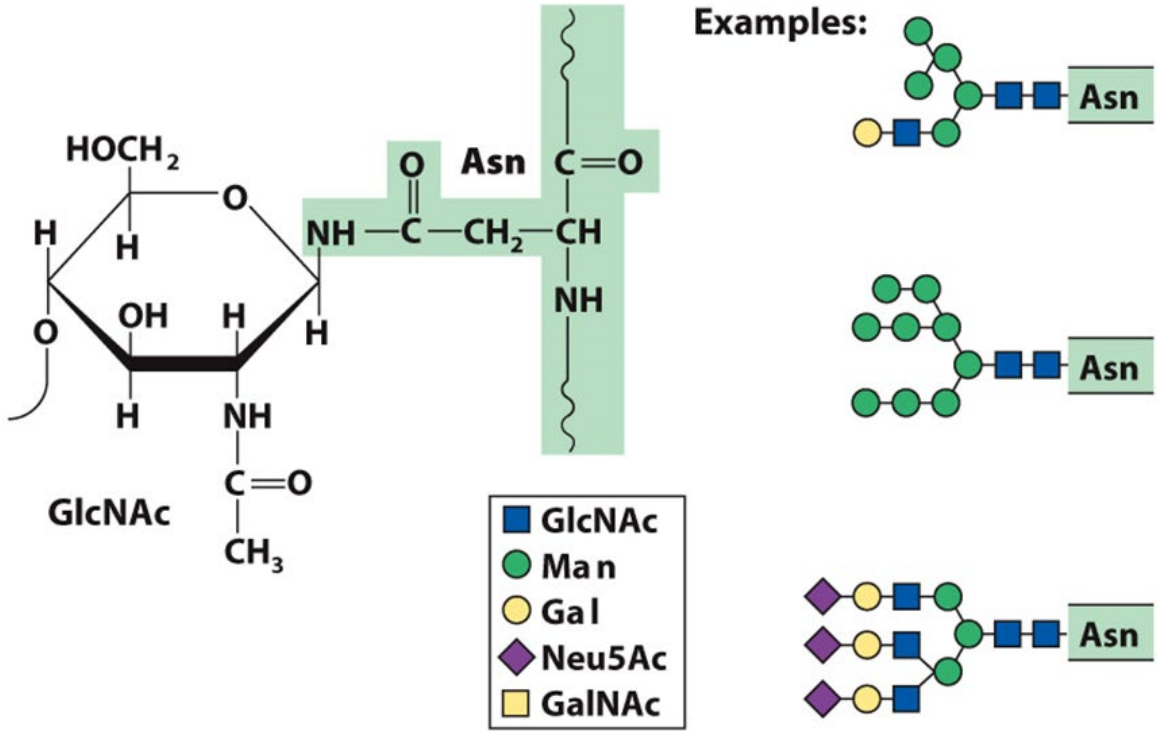


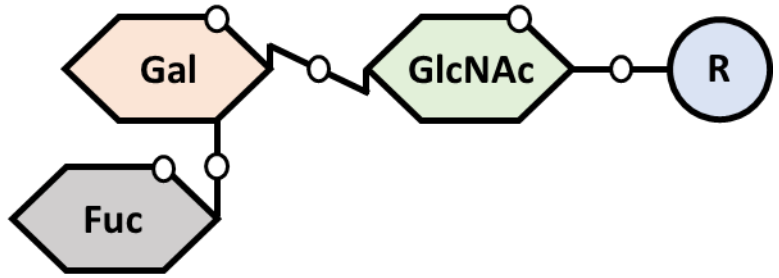
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Glycolipids

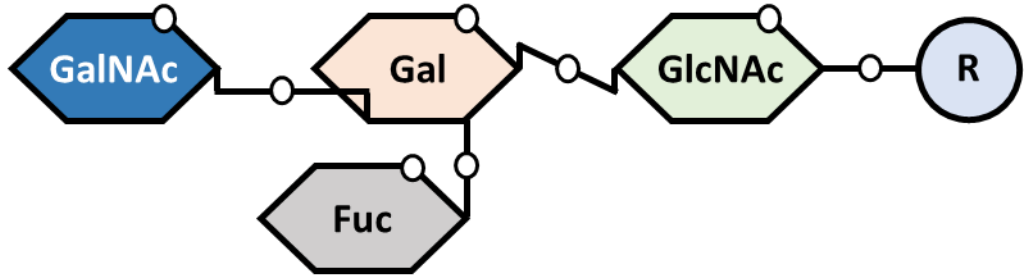
- Lipids with covalently bound oligosaccharides
- Lipopolysaccharide is a major component of the outer membrane of Gram-negative bacteria
- ABO blood-group antigens are oligosaccharide structures present on cell-surface glycoproteins and glycolipids

ABO blood types

Type 0



Type A



Type B

